WEAK-TO-STRONG GENERALIZATION: ELICITING STRONG CAPABILITIES WITH WEAK SUPERVISION

Trustworthy AI - Midterm Paper Presentation - NYCU

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Venue/Year: ICML 2024 (PMLR 235)

Target task: Can a weak supervisor elicit the full capability of a much stronger model?

Research Question and Motivation

Research question:

Can weak supervision elicit everything a much stronger model knows?

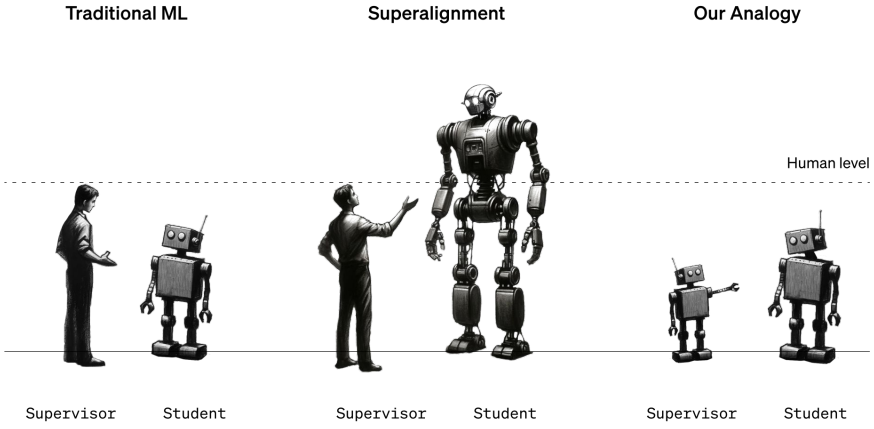
Motivation:

Alignment methods such as RLHF assume humans can judge model outputs.

Future models may outperform their human supervisors: oversight becomes weak.

Gap: no empirical way to study this today, since no model is superhuman yet

Idea: replace the weak human with a weak model, and study the analogy



Burns et al. (2024), Fig. 1

Related Work and Positioning

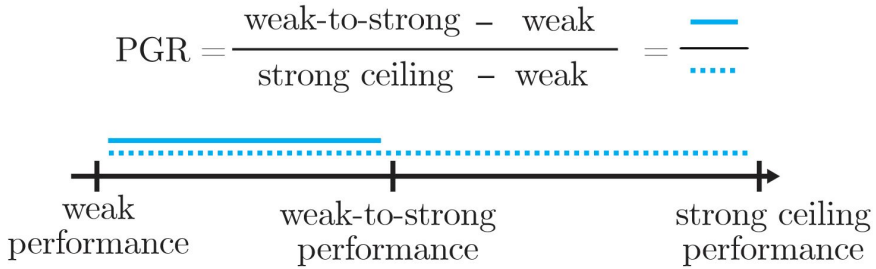
Two closely related studies:

Reed et al. (Google) (2014), arXiv:1412.6596:
Learning with noisy labels: treats label errors as noise to be resisted.

Bowman et al. (Anthropic) (2022), arXiv:2211.03540:
Scalable oversight: makes the supervisor stronger, e.g. humans assisted by AI.

Difference: weak-model errors are systematic, and the student is far more capable than its labels

Trade-off: a fast, repeatable testbed, but only an analogy of human supervision



Burns et al. (2024)

Method: The Core Idea

Protocol:

1. train a weak model on ground truth
2. its predictions become weak labels
3. fine-tune a strong pretrained model on them
4. compare with the strong model trained on ground truth (ceiling)

Auxiliary confidence loss (App. A.4):

$$L_{\text{conf}}(f) = (1-\alpha)\cdot \text{CE}(f(x), f_w(x)) + \alpha \cdot \text{CE}(f(x), \hat{f}_t(x))$$

f_w : weak label

f^_t : student's own hardened prediction (threshold t)

α : warmed up from 0 to 0.75

Naive baseline: the student imitates the weak labels with cross-entropy

Key method: an auxiliary confidence loss lets the student trust its own confident predictions

What is new: weak supervision as a test of eliciting latent knowledge, not of cleaning noisy labels

Main Findings and Supporting Evidence

Setting:

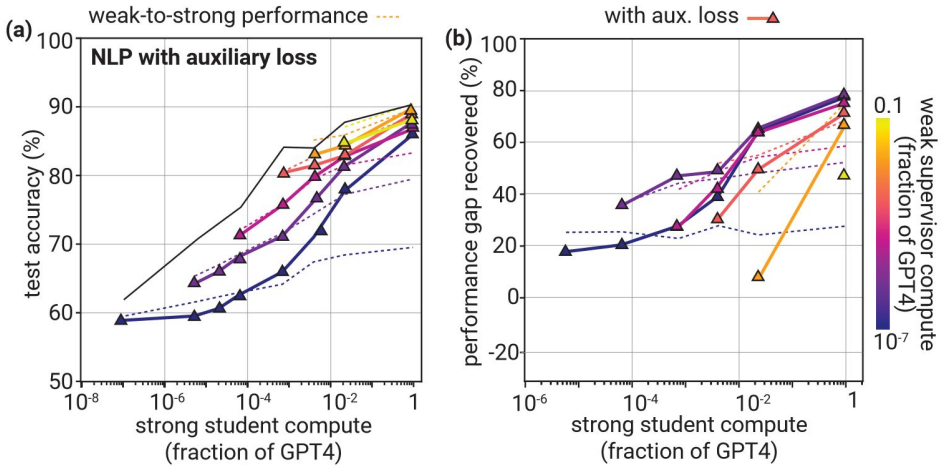
GPT-4-family models; 22 NLP tasks, chess puzzles, ChatGPT reward modelling

Finding:

- Naive students beat their supervisors almost everywhere.
- With the confidence loss: median PGR from about 25 % to nearly 80 % (largest gap, NLP)
- But reward modelling recovers only about 10%; students overfit to weak errors

Scope:

Weak-to-strong gains are real but method- and task-dependent



Burns et al. (2024), Fig. 5

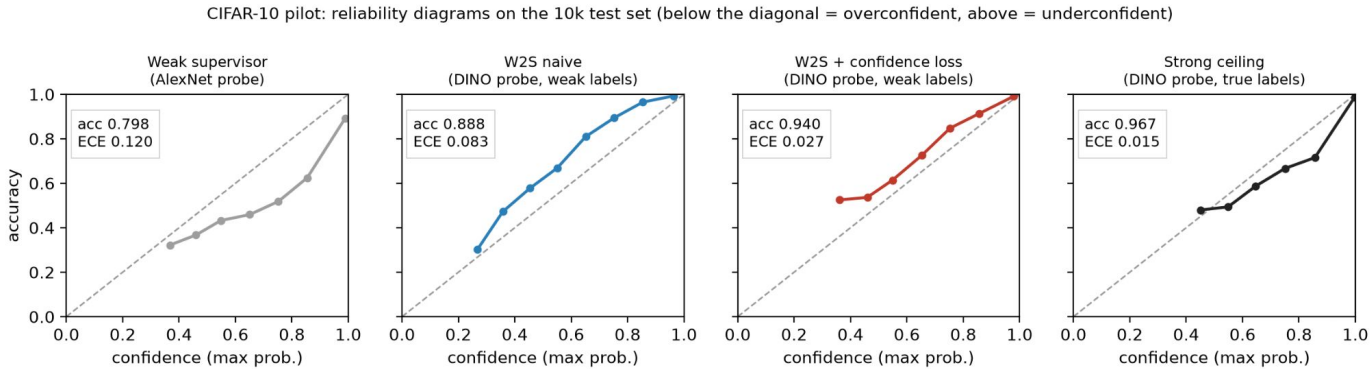
Critical Review: Assumptions, Limits, and Implications

What is convincing?

- Strength: a measurable, reproducible proxy for superhuman oversight
- Evidence: positive PGR across three domains and many model sizes.

Where might it fail?

- Limitation: judged by accuracy and PGR only; calibration is never measured
- Consequence: in the real setting no labels exist to catch confident mistakes



My own work: AlexNet supervising DINO ViT-B/8 linear probe on CIFAR-10.

Extension, Further Directions, and Takeaway

One concrete extension of my own: Uncertainty-aware weak-to-strong generalization

- Measure calibration and uncertainty, not only PGR: ECE, error detection, conformal coverage
- Weight the confidence term per example by uncertainty (ensemble agreement, supervisor entropy) instead of a fixed α
- Fix conformal sets calibrated on weak labels (pilot: 99.3 % coverage for a 90 % target, sets 66 % too large)

Further direction:

Label-free estimation of a student's reliability, an open question the authors pose (Sec. 6)

Takeaway:

Weak supervision can elicit strong capabilities, but trust needs calibrated uncertainty, not accuracy alone